Q1.A payment system supports different payment methods.

```
CLASS Payment
    METHOD process()
        PRINT "Generic Payment"

    METHOD process(int amount)
        PRINT "Payment: " + amount
END CLASS

CLASS CreditCardPayment EXTENDS Payment

    METHOD process()
        PRINT "Credit Card"

    METHOD process(double amount)
        PRINT "Credit Card: " + amount
END CLASS

Payment p = NEW CreditCardPayment()
p.process()
p.process(500)
```

Q2.What is the output of the following pseudocode?

```
CLASS Parent
    int value = 10

    METHOD show()
        PRINT value
END CLASS

CLASS Child EXTENDS Parent
    int value = 20

    METHOD show()
        PRINT value
END CLASS

Parent p = NEW Child()
PRINT p.value
p.show()
```

Q3.What is the output of the following pseudocode?

```
CLASS Parent

    Parent()
        show()

    METHOD show()
        PRINT "Parent"
END CLASS

CLASS Child EXTENDS Parent
    int x = 20

    Child()
        PRINT x

    METHOD show()
        PRINT x
END CLASS

Child c = NEW Child()
```

Q4.What is the output of the following pseudocode?

```
CLASS Parent

    STATIC METHOD display()
        PRINT "Parent"
END CLASS


CLASS Child EXTENDS Parent

    STATIC METHOD display()
        PRINT "Child"
END CLASS


Parent p = NEW Child()
p.display()
```

Q5. What is the output of the following pseudocode?

```
CLASS Animal
END CLASS

CLASS Dog EXTENDS Animal
END CLASS

CLASS Parent

   METHOD getAnimal()
      RETURN NEW Animal()
END CLASS

CLASS Child EXTENDS Parent

   METHOD getAnimal()
      RETURN NEW Dog()
END CLASS
```

Q6.What is the output of the following pseudocode?

```
CLASS Parent

    PRIVATE METHOD show()
        PRINT "Parent"

    METHOD execute()
        show()
END CLASS

CLASS Child EXTENDS Parent

    METHOD show()
        PRINT "Child"
END CLASS

Child c = NEW Child()
c.execute()
```

Q7.What is the output of the following pseudocode?

```
CLASS Employee

    METHOD calculate()
        PRINT "Employee"

    METHOD process()
        calculate()
END CLASS

CLASS Manager EXTENDS Employee

    METHOD calculate()
        PRINT "Manager"

    METHOD process()
        super.process()
        calculate()
END CLASS

Employee e = NEW Manager()
e.process()
```

Q8.What is the output of the following pseudocode?

```
CLASS A

    A()
        PRINT "A"

    A(int x)
        PRINT "A1"
END CLASS

CLASS B EXTENDS A

    B()
        this(10)
        PRINT "B"

    B(int x)
        super(x)
        PRINT "B1"
END CLASS

B obj = NEW B()
```

Q9.What is the output of the following pseudocode?

```
INTERFACE A
    DEFAULT METHOD show()
        PRINT "A"
END INTERFACE

INTERFACE B
    DEFAULT METHOD show()
        PRINT "B"
END INTERFACE

CLASS C IMPLEMENTS A, B
END CLASS
```

Q10.What is the output of the following pseudocode?

```
ABSTRACT CLASS Report

    Report()
        PRINT "Report Constructor"

    ABSTRACT METHOD generate()
END CLASS

CLASS SalesReport EXTENDS Report

    SalesReport()
        PRINT "Sales Constructor"

    METHOD generate()
        PRINT "Sales Report"
END CLASS

Report r = NEW SalesReport()
r.generate()
```

A.
Sales Constructor
Sales Report

B.
Report Constructor
Sales Constructor
Sales Report

C.
Report Constructor
Sales Report

D.
Compilation error because abstract classes cannot have constructors

Q11.What is the output of the following pseudocode?

```
CLASS Printer

   METHOD print(int x)
      PRINT "int"

METHOD print(Integer  x)
      PRINT “Integer"


   METHOD print(double x)
      PRINT "double"

   METHOD print(String x)
      PRINT "String"
END CLASS

Printer p = NEW Printer()

byte b = 10
p.print(b)
```

Q12.What is the output of the following pseudocode?

```
CLASS Parent

    METHOD calculate(int x)
        PRINT "Parent int"
END CLASS

CLASS Child EXTENDS Parent

    METHOD calculate(double x)
        PRINT "Child double"
END CLASS

Parent p = NEW Child()
p.calculate(10)
```

Q13.What is the output of the following pseudocode?

```
CLASS A
    METHOD show()
        PRINT "A"
END CLASS

CLASS B EXTENDS A
    METHOD show()
        PRINT "B"
END CLASS

CLASS C EXTENDS B
    METHOD show()
        PRINT "C"
END CLASS

A obj = NEW C()

obj.show()
```

Q14.What is the output of the following pseudocode?

```
CLASS Employee
    METHOD work()
        PRINT "Employee"

    METHOD report()
        PRINT "Report"
END CLASS

CLASS Manager EXTENDS Employee

    METHOD work()
        PRINT "Manager"

    METHOD approve()
        PRINT "Approved"
END CLASS

Employee e = NEW Manager()
e.work()
e.report()
```

Q15. What is the output of the following pseudocode?

```
CLASS Animal
END CLASS

CLASS Dog EXTENDS Animal
    METHOD bark()
        PRINT "Bark"
END CLASS

Animal a = NEW Dog()

Dog d = (Dog) a

d.bark()
```

Q16.What is the output of the following pseudocode?

```
CLASS Parent
   STATIC int count = 10
END CLASS

CLASS Child EXTENDS Parent
   STATIC int count = 20
END CLASS

Parent p = NEW Child()
Child c = NEW Child()

PRINT p.count
PRINT c.count
```

Q.17 What is the output of the following pseudocode?

```
INTERFACE Notification
    METHOD send()
END INTERFACE

CLASS Email IMPLEMENTS Notification
    METHOD send()
        PRINT "Email"
END CLASS

CLASS SMS IMPLEMENTS Notification
    METHOD send()
        PRINT "SMS"
END CLASS

Notification n = NEW SMS()
n.send()
```

Q18.What is the output of the following pseudocode?

```
CLASS Parent

    METHOD show(Object o)
        PRINT "Parent Object"

    METHOD show(String s)
        PRINT "Parent String"
END CLASS

CLASS Child EXTENDS Parent

    METHOD show(Object o)
        PRINT "Child Object"

    METHOD show(Integer i)
        PRINT "Child Integer"
END CLASS

Parent p = NEW Child()
p.show(null)
```

Q19.What is the output of the following pseudocode?

```
INTERFACE A
   METHOD show()
END INTERFACE

ABSTRACT CLASS B IMPLEMENTS A

   METHOD show()
      PRINT "B"
END CLASS

CLASS C EXTENDS B

   METHOD show()
      PRINT "C"
END CLASS

A obj = NEW C()
obj.show()
```

Q20. What is the output of the following pseudocode?

```
CLASS A
    int x = 10
    METHOD display()
        PRINT x
    METHOD process()
        display()
END CLASS

CLASS B EXTENDS A
    int x = 20
    METHOD display()
        PRINT x
    METHOD process()
        super.process()
        PRINT super.x
END CLASS
A obj = NEW B()
obj.process()
```

Q21.What is the output of the following pseudocode?

```
CLASS A

    int value = 5

    A()
        display()

    METHOD display()
        PRINT value
END CLASS
```

```
CLASS B EXTENDS A

    int value = 10

    B()
        PRINT value

    METHOD display()
        PRINT value
END CLASS
```

```
CLASS C EXTENDS B

    int value = 20

    C()
        PRINT value

    METHOD display()
        PRINT value
END CLASS


A obj = NEW C()
```

Q22.What is the output of the following pseudocode?

```
CLASS Parent
    STATIC
        PRINT "P-Static"
    Parent()
        PRINT "P-Constructor"
END CLASS

CLASS Child EXTENDS Parent
    STATIC
        PRINT "C-Static"
    Child()
        PRINT "C-Constructor"
END CLASS

Child c1 = NEW Child()
Child c2 = NEW Child()
```

A.

P-Static
C-Static
P-Constructor
C-Constructor
P-Constructor
C-Constructor

B.

P-Static
P-Constructor
C-Static
C-Constructor
P-Constructor
C-Constructor

C.

C-Static
P-Static
P-Constructor
C-Constructor
P-Constructor
C-Constructor

D.

P-Static
C-Static
C-Constructor
P-Constructor
C-Constructor
P-Constructor